CSE476-Large Language Models
EcoRisk LLM

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ABSTRACT

Wildfires are a major threat to the environment, especially in regions like Antalya. This project introduces EcoRisk LLM, an automated system that generates wildfire risk reports. The system uses a Large Language Model (Llama-3) combined with Retrieval-Augmented Generation (RAG) technology.

Unlike standard AI models, EcoRisk LLM connects to a scientific knowledge base using a FAISS vector database. This allows the model to retrieve ecological facts about vegetation, weather, and topography before writing a report. Our results show that the system achieves 100% factual consistency, meaning it uses numerical data without any mistakes. By turning raw data into human-readable text, EcoRisk LLM helps experts and decision-makers understand wildfire risks more easily and accurately.

PROBLEM DEFINITION AND MOTIVATION

Wildfire data (weather, vegetation, topography) is complex and hard for non-experts to interpret quickly. Additionally, standard AI models often "hallucinate" or misinterpret numerical data, which can lead to dangerous errors in risk assessment. Our motivation is to turn raw data into reliable, expert-level reports. By using **RAG technology**, we prevent AI errors and ensure that every report is based on scientific facts. This project aims to help decision-makers in Antalya take faster, more accurate actions to prevent forest fires.

DATASET AND DATA SOURCE

The dataset for this project was originally collected for our **senior graduation project**. It covers the **Antalya region** and contains over 5,000 data points from **Google Earth Engine (GEE)**. Having this data already available allowed us to focus on the Natural Language Processing (NLP) pipeline for this class.

Each data point includes:

- **Geography:** Altitude and Slope.
- **Meteorology:** Temperature and Wind Speed.
- **Vegetation:** NDVI and NDWI values.
- **Fire History:** Labels for historical fire occurrences.

For our testing, we selected **20 balanced samples** (10 fires, 10 no-fires) and **3 edge-case scenarios** to evaluate how well the LLM interprets the physical correlations in the data.

LLM METHODOLOGY

Our methodology focuses on combining raw data with expert knowledge using a **Real RAG (Retrieval-Augmented Generation)** pipeline.

5.1. Model and Infrastructure: We used the **Llama-3.1-8B** model (accessible via the Groq API). This model provides a high-speed and high-quality natural language generation. Instead of fine-tuning, we used **Prompt Engineering** and **RAG** to guide the model's reasoning.

5.2. RAG Pipeline (FAISS): We built a knowledge base of 26 ecological rules. We used a **FAISS** vector database and **Sentence Transformers** to perform semantic searches. For every input, the system finds the 3 most relevant scientific facts and injects them into the prompt. This ensures the model explains the "why" behind the fire risk.

5.3. Prompting Strategy: We used a **3-Shot Prompting** strategy. This means we gave the model three examples of how to analyze data and write a report. We also used a strict system command to force the model to stay objective and always include a specific risk level (Low, Medium, High, or Very High).

5.4. Evaluation Setup: We developed a unique evaluation framework with three metrics:

1. **Exact-Match:** A custom algorithm to verify if the LLM correctly uses the numerical data without making mistakes.
2. **Classification Metrics:** Accuracy and F1-Score to check if the LLM's risk label matches historical data.
3. **ROUGE:** A standard NLP metric to measure the linguistic richness of the generated text.

EXPERIMENTS AND RESULTS

We tested our system on 20 balanced records and 3 special edge-case scenarios. The results are summarized below:

6.1 Quantitative Results

METRIC	SCORE
Exact-Match Factual Consistency	**100%**
Classification Accuracy	**55%**
Recall (Fire Detection)	**100%**

Exact-Match: The system achieved a perfect 100% score. This means the model never hallucinated or changed the input numbers (temperature, wind, etc.) in the final report.

Accuracy: The accuracy was 55%. This is because the model is "conservative"—it often identifies a risk even if a fire did not happen historically. However, the **Recall is 100%**, meaning the system never missed a real fire event.

6.2. Qualitative Analysis: The reports generated by the LLM were scientifically rich. By using RAG, the model correctly identified the **"Chimney Effect"** on steep slopes and explained how **NDWI (water stress)** increases the risk of fire in dense forests. The model moved beyond simple labels and provided deep ecological reasoning.

6.3. Edge-Case Performance: In our edge-case tests, the model successfully identified that a "Very High Temperature" does not mean a "High Risk" if there is **no fuel** (NDVI < 0.15). This demonstrates that the RAG pipeline effectively guides the model's logical reasoning.

DISCUSSIONS AND LIMITATIONS

The project proves that a **Real RAG pipeline** is much more reliable than a standard LLM. By using FAISS, we successfully combined scientific ecological knowledge with real-time data. The 100% Exact-Match score is our biggest success, showing that the model is safe for professional use because it does not change numerical facts. The system acts like an expert assistant rather than just a simple calculator.

Limitations:

1. **API Rate Limits:** We used the Groq API, which has a "Tokens Per Day" limit. This prevented us from testing thousands of records at once.

2. **Sample Size:** Due to these API limits and time constraints, we evaluated 20 records. While these were carefully selected (stratified), a larger dataset would provide even more proof of success.
3. **Parsing Complexity:** Since the LLM produces free text, converting "Moderate Risk" or "Medium Danger" into a binary (0 or 1) label can sometimes be difficult. A more advanced natural language parser could improve accuracy.
4. **Static Knowledge Base:** Our knowledge base has 26 rules. Adding hundreds of more detailed scientific papers to the FAISS database would improve the depth of the reports.

CONCLUSION

In conclusion, **EcoRisk LLM** successfully demonstrates how Large Language Models can be used for professional ecological risk assessment. By integrating **Real RAG** and a **FAISS vector database**, we solved the problem of AI hallucinations and ensured 100% numerical accuracy. The system does not only predict risk but also explains the scientific reasons behind it. This framework is scalable and can be used for different regions or environmental disasters in the future. Our project proves that AI, when guided by scientific knowledge, can be a powerful tool for environmental protection.